TEST OUTPUTS:

FUNCTIONAL TESTS:

```
n_number": 5433, "name": "PotionCo", "type": "sell", "quantity": 1}, "5434": {"transaction_number": 5434, "name": "PotionCo", "type": "buy", "quantity": 1}, "5435": {"transaction_number": 5435, "name": "Sword
Co", "type": "sell", "quantity": 8}, "5436": {"transaction_number": 5436, "name": "GameStart", "type": "buy", "quantity": 9}, "5437": {"transaction_number": 5437, "name": "GameStart", "type": "buy", "quantity
": 2}, "5438": {"transaction_number": 5438, "name": "ArrowCo", "type": "buy", "quantity": 6}, "5439": {"transaction_number": 5439, "name": "ArrowCo", "type": "buy", "quantity": 9}, "5440": {"transaction_numbe
r": 5440, "name": "ShieldCo", "type": "sell", "quantity": 8}, "5441": {"transaction_number": 5441, "name": "BowCo", "type": "sell", "quantity": 8}, "5442": {"transaction_number": 5442, "name": "RottenFishCo",
 "type": "buy", "quantity": 10}, "5443": {"transaction_number": 5443, "name": "ShieldCo", "type": "buy", "quantity": 7}, "5444": {"transaction_number": 5444, "name": "BoarCo", "type": "buy", "quantity": 3}, "
5445": {"transaction_number": 5445, "name": "GameStart", "type": "sell", "quantity": 7}, "5446": {"transaction_number": 5446, "name": "BoarCo", "type": "buy", "quantity": 9}, "5447": {"transaction_number": 54
47, "name": "ShieldCo", "type": "buy", "quantity": 9}, "5448": {"transaction_number": 5448, "name": "ArrowCo", "type": "sell", "quantity": 5}, "5449": {"transaction_number": 5449, "name": "PotionCo", "type":
"sell", "quantity": 8}, "5450": {"transaction_number": 5450, "name": "BowCo", "type": "buy", "quantity": 3}, "5451": {"transaction_number": 5451, "name": "BoarCo", "type": "buy", "quantity": 7}, "5452": {"tra
nsaction_number": 5452, "name": "ArrowCo", "type": "sell", "quantity": 1}, "5453": {"transaction_number": 5453, "name": "ShieldCo", "type": "buy", "quantity": 9}, "5454": {"transaction_number": 5454, "name":
"SwordCo", "type": "buy", "quantity": 5}, "5455": {"transaction_number": 5455, "name": "PotionCo", "type": "buy", "quantity": 10}, "5456": {"transaction_number": 5456, "name": "PotionCo", "type": "sell", "qua
ntity": 1}, "5457": {"transaction_number": 5457, "name": "ArrowCo", "type": "sell", "quantity": 2}, "5458": {"transaction_number": 5458, "name": "GameStart", "type": "sell", "quantity": 3}, "5459": {"transact
ion_number": 5459, "name": "PotionCo", "type": "sell", "quantity": 1}, "5460": {"transaction_number": 5460, "name": "ShieldCo", "type": "buy", "quantity": 7}, "5461": {"transaction_number": 5461, "name": "Bow
Co", "type": "buy", "quantity": 10}, "5462": {"transaction_number": 5462, "name": "GameStart", "type": "buy", "quantity": 1}, "5463": {"transaction_number": 5463, "name": "GameStart", "type": "sell", "quantit
y": 1}, "5464": {"transaction_number": 5464, "name": "PotionCo", "type": "buy", "quantity": 7}, "5465": {"transaction_number": 5465, "name": "ArrowCo", "type": "sell", "quantity": 3}, "5466": {"transaction_nu
mber": 5466, "name": "BowCo", "type": "buy", "quantity": 10}, "5467": {"transaction_number": 5467, "name": "PotionCo", "type": "buy", "quantity": 3}, "5468": {"transaction_number": 5468, "name": "BoarCo", "ty
pe": "sell", "quantity": 5}, "5469": {"transaction_number": 5469, "name": "BowCo", "type": "sell", "quantity": 4}, "5470": {"transaction_number": 5470, "name": "GameStart", "type": "buy", "quantity": 9}, "547
1": {"transaction_number": 5471, "name": "GameStart", "type": "sell", "quantity": 7}, "5472": {"transaction_number": 5472, "name": "ShieldCo", "type": "sell", "quantity": 2}, "5473": {"transaction_number": 54
73, "name": "BoarCo", "type": "buy", "quantity": 1}, "5474": {"transaction_number": 5474, "name": "PotionCo", "type": "buy", "quantity": 5}, "5475": {"transaction_number": 5475, "name": "BowCo", "type": "buy"
, "quantity": 6}}}
frontend_service-1  | 172.18.0.7 - - [08/May/2025 03:42:19] "GET /orders HTTP/1.1" 200 -
order_service_1-1   | 172.18.0.7 - - [08/May/2025 03:42:19] "GET /orders/999 HTTP/1.1" 200 -
order_service_2-1   | 172.18.0.7 - - [08/May/2025 03:42:19] "GET /orders/999 HTTP/1.1" 200 -
order_service_3-1   | 172.18.0.7 - - [08/May/2025 03:42:19] "GET /orders/999 HTTP/1.1" 200 -
order_service_3-1   | [INFO] Leader updated to 3
order_service_3-1   | [SYNC] Trying to sync with: ['http://order_service_1:6001', 'http://order_service_2:6002', 'http://order_service_3:6003']
order_service_3-1   | [SYNC] Synced 0 txns from http://order_service_1:6001
order_service_3-1   | {'name': 'INVALIDDD', 'quantity': 10, 'type': 'buy', 'followers': ['http://order_service_1:6001', 'http://order_service_2:6002']}
order_service_3-1   | no validation error
frontend_service-1  | 172.18.0.7 - - [08/May/2025 03:42:19] "GET /orders HTTP/1.1" 200 -
catelog_service-1   | 172.18.0.3 - - [08/May/2025 03:42:19] "POST /update/INVALIDDD/-10 HTTP/1.1" 404 -
frontend_service-1  | 172.18.0.7 - - [08/May/2025 03:42:19] "POST /orders HTTP/1.1" 400 -
catelog_service-1   | [INIT] Catalog loaded into memory: {'GameStart': {'price': 15.99, 'quantity': 120}, 'RottenFishCo': {'price': 5.49, 'quantity': 293}, 'BoarCo': {'price': 9.99, 'quantity': 65}, 'MenhirCo
': {'price': 12.75, 'quantity': 0}, 'SwordCo': {'price': 25.0, 'quantity': 168}, 'ShieldCo': {'price': 30.0, 'quantity': 32}, 'BowCo': {'price': 35.0, 'quantity': 70}, 'ArrowCo': {'price': 2.5, 'quantity': 11
6}, 'PotionCo': {'price': 10.0, 'quantity': 162}, 'HelmetCo': {'price': 15.0, 'quantity': 0}}
order_service_3-1   | 172.18.0.6 - - [08/May/2025 03:42:19] "POST /orders HTTP/1.1" 404 -
catelog_service-1   | Catalog Service running on 0.0.0.0:5002
catelog_service-1   | reached catalog service -   /update/INVALIDDD/-10
frontend_service-1  | 172.18.0.7 - - [08/May/2025 03:42:19] "POST /orders HTTP/1.1" 404 -
catelog_service-1   | reaching inside update -  /update/INVALIDDD/-10
frontend_service-1  | 172.18.0.7 - - [08/May/2025 03:42:19] "POST /orders HTTP/1.1" 404 -
order_service_3-1   | 172.18.0.6 - - [08/May/2025 03:42:19] "POST /orders HTTP/1.1" 400 -
order_service_3-1   | {'name': 'GameStart', 'quantity': -10, 'type': 'buy', 'followers': ['http://order_service_1:6001', 'http://order_service_2:6002']}
frontend_service-1  | 172.18.0.7 - - [08/May/2025 03:42:19] "POST /orders HTTP/1.1" 400 -
frontend_service-1  | 172.18.0.7 - - [08/May/2025 03:42:19] "POST /orders HTTP/1.1" 400 -
order_service_3-1   | {'name': 'GameStart', 'quantity': 10, 'type': 'invalid_type', 'followers': ['http://order_service_1:6001', 'http://order_service_2:6002']}
order_service_3-1   | 172.18.0.6 - - [08/May/2025 03:42:19] "POST /orders HTTP/1.1" 400 -
frontend_service-1  | 172.18.0.7 - - [08/May/2025 03:42:19] "POST /orders HTTP/1.1" 400 -
frontend_service-1  | 172.18.0.7 - - [08/May/2025 03:42:19] "POST /orders HTTP/1.1" 400 -
order_service_3-1   | {'name': 'GameStart', 'quantity': 999999, 'type': 'buy', 'followers': ['http://order_service_1:6001', 'http://order_service_2:6002']}
order_service_3-1   | no validation error
catelog_service-1   | 172.18.0.3 - - [08/May/2025 03:42:19] "POST /update/GameStart/-999999 HTTP/1.1" 400 -
catelog_service-1   | reached catalog service -   /update/GameStart/-999999
catelog_service-1   | reaching inside update -  /update/GameStart/-999999
order_service_3-1   | 172.18.0.6 - - [08/May/2025 03:42:19] "POST /orders HTTP/1.1" 400 -
frontend_service-1  | 172.18.0.7 - - [08/May/2025 03:42:19] "POST /orders HTTP/1.1" 400 -
frontend_service-1  | 172.18.0.7 - - [08/May/2025 03:42:19] "POST /orders HTTP/1.1" 400 -
test_runner-1       | ..........
test_runner-1       | ----------------------------------------------------------------------
test_runner-1       | Ran 10 tests in 1.078s
test_runner-1       |
test_runner-1       | OK
test_runner-1       | --------
test_runner-1       | <Response [200]>
```

Unit tests:

```
(3.10.13) →  spring25-lab3-spring25-lab3-shireenmeher git:(main) ✗ python3 src/frontend_service/src/test_main.py
printing fe
[INFO] Sorted replicas: [{'id': 3, 'url': 'http://order_service_3:6003'}, {'id': 2, 'url': 'http://order_service_2:6002'}
, {'id': 1, 'url': 'http://order_service_1:6001'}]
 * Serving Flask app 'test_main'
 * Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
 * Running on http://127.0.0.1:5002
Press CTRL+C to quit
.[ERROR] Catalog request failed: service down
Traceback (most recent call last):
  File "/Users/shireenmeher/Documents/compsci677/lab3/spring25-lab3-spring25-lab3-shireenmeher/src/frontend_service/src/m
ain.py", line 184, in do_GET
    cat_res = requests.get(f"{CATALOG_SERVICE_URL}/stocks/{stock_name}")
  File "/Users/shireenmeher/.pyenv/versions/3.10.13/lib/python3.10/unittest/mock.py", line 1114, in __call__
    return self._mock_call(*args, **kwargs)
  File "/Users/shireenmeher/.pyenv/versions/3.10.13/lib/python3.10/unittest/mock.py", line 1118, in _mock_call
    return self._execute_mock_call(*args, **kwargs)
  File "/Users/shireenmeher/.pyenv/versions/3.10.13/lib/python3.10/unittest/mock.py", line 1173, in _execute_mock_call
    raise effect
Exception: service down
..[ERROR] Invalid JSON body: Expecting value: line 1 column 1 (char 0)
.[INFO] Sorted replicas: [{'id': 3, 'url': 'http://order_service_3:6003'}, {'id': 2, 'url': 'http://order_service_2:6002'
}, {'id': 1, 'url': 'http://order_service_1:6001'}]
raching notify all replicas
[LEADER] Selected replica 3
reached here you predicted correct
.reached here you predicted correct
Test response: {"data": {"transaction_number": 123}}{"data": {"transaction_number": 123}}
.
----------------------------------------------------------------------
Ran 6 tests in 0.019s

OK
(3.10.13) →  spring25-lab3-spring25-lab3-shireenmeher git:(main) ✗
Ln 177, Col 1   Spaces: 4   UTF-8   LF   {} Python   3.10.13 ('3.10.13')
```

The error msgs shown are because of testing of a failure condition.

```
OK
(3.10.13) → spring25-lab3-spring25-lab3-shireenmeher git:(main) ✗ python3 src/order_service/src/test_main.py
this is a print statement
{'name': 'GameStart', 'quantity': 2000, 'type': 'buy'}
no validation error
....{'name': '', 'quantity': 0, 'type': 'hold'}
.set leader working
[INFO] Leader updated to 3
..{'name': 'GameStart', 'quantity': 2, 'type': 'buy'}
no validation error
update successful
update propagation successful
.{'name': 'BoarCo', 'quantity': 5, 'type': 'sell'}
no validation error
update successful
update propagation successful
.
----------------------------------------------------------------------
Ran 9 tests in 0.007s

OK
(3.10.13) → spring25-lab3-spring25-lab3-shireenmeher git:(main) ✗
```

```
(3.10.13) → spring25-lab3-spring25-lab3-shireenmeher git:(main) ✗ python3 src/catelog_service/src/test_main.py
...
----------------------------------------------------------------------
Ran 3 tests in 0.001s

OK
(3.10.13) → spring25-lab3-spring25-lab3-shireenmeher git:(main) ✗
Not Committed Yet  Ln 20, Col 12  Spaces: 4  UTF-8  LF  {} Python  3.10.13 ('3.10.13')
```